Absolute stability from inverse folding

Predicting absolute protein folding stability using generative models

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Can a model trained to design sequences also estimate how much a protein resists unfolding?

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The question is harder than ranking point mutations

Established task: $\Delta\Delta G$

Compare a wild type and a nearby variant, usually in the same fold.

Many energetic contributions and some experimental-condition effects cancel.

This task: absolute ΔG

Compare unrelated sequences and folds at a specified temperature and pH.

Must implicitly account for folded and unfolded states, solvent, and size.

Contribution: reuse a pretrained inverse-folding model, then fit only a linear conversion to physical units.

Introduction; Results 2.1. ΔG here denotes free energy of unfolding.

What exactly is being predicted?

$$\Delta G_{\text{unfold}} = G_U - G_F = RT \log \frac{p_F}{p_U}.$$

Positive ΔG means the folded state is thermodynamically favored.

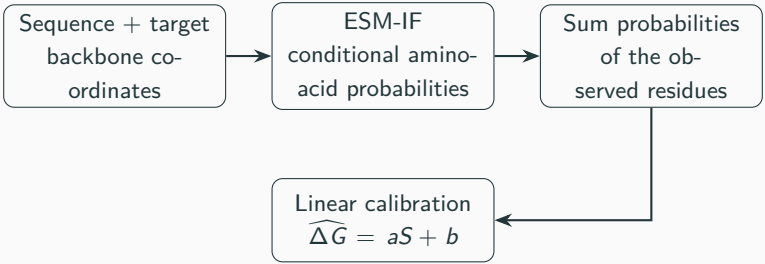
Illustrative two-state example at 25°C:
$RT \simeq 0.592$ kcal/mol. A stability of 4 kcal/mol gives $p_F/p_U \simeq e^{4/0.592} \simeq 860$.

An error of 1.4 kcal/mol corresponds to about a 10.6-fold error in this equilibrium ratio.

Prediction target: ambient-condition equilibrium stability; neither melting temperature nor a folding rate.

Definition: Introduction. Equilibrium calculation is a teaching example, not a measured result.

From a backbone to a calibrated stability estimate



ESM-IF: geometric backbone encoder + autoregressive transformer decoder. The structure and preceding native residues condition each prediction.

The pretrained network is reused; stability measurements calibrate two parameters.

Introduction; Methods 4.2–4.3. Model: esm_if1_gvp4_t16_142M_UR50.

The key score sums probabilities, not log probabilities

For sequence $s = (s_1, \ldots, s_N)$ and backbone X:

$$p_i = \Pr_{\mathrm{ESM-IF}}(s_i \mid X, s_{<i}), \qquad S = \sum_{i=1}^{N} p_i.$$

1. Softmax the decoder logits over the 20 amino acids.
2. Select the probability assigned to the actual residue at each position.
3. Add these probabilities across the protein.

This is *not* the sequence log likelihood $\sum_i \log p_i$, and no Boltzmann derivation is asserted. It is an empirical sequence–structure compatibility score with an explicit size dependence.

Equation 1; Methods 4.3; author SI notebook, wt_ll().

A concrete example: observed residues become a scalar

Position	1	2	3	4
Observed residue	V	A	I	G
Model probability	.60	.25	.50	.10

$$S = 0.60 + 0.25 + 0.50 + 0.10 = 1.45.$$

For a realistic-length *illustrative* protein with $S = 40$, the author notebook's fitted conversion gives

$$\widehat{\Delta G} = 0.10687 \times 40 + 0.51921 \simeq 4.79 \ \mathrm{kcal/mol}.$$

The four-residue example only teaches the summation; such a peptide lies outside the benchmark. Calibration is not a universal physical constant.

Author SI notebook: linear_curve() and saved fitted-parameter output. Numbers in examples are synthetic.

The benchmark makes the thermodynamic target more coherent

Source	Proteins	Length	Conditions
Maxwell et al. (2005)	26	61–151 aa	25°C, pH 7.0
Tsuboyama et al. (2023)	239	40–72 aa	∼25°C, pH 7.4

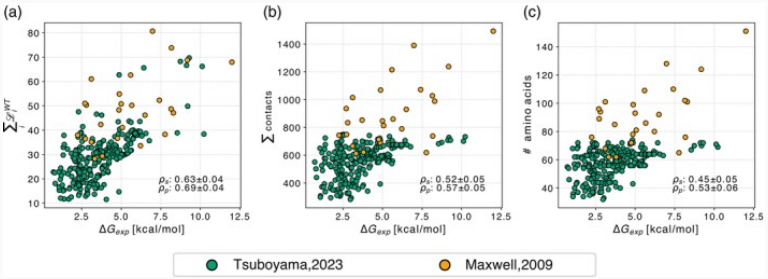
265 natural proteins; experimental ΔG roughly 1–12 kcal/mol.

Designed proteins excluded from the megascale set; obligate homodimers and VlsE excluded from the curated set. Focus: equilibrium two-state folding.

Interpretation: many measurements, but most proteins are short and come from one experimental platform.

Results 2.1; Methods 4.1. Component-specific length ranges are reported as stated in the text.

Sequence–structure compatibility beats size alone

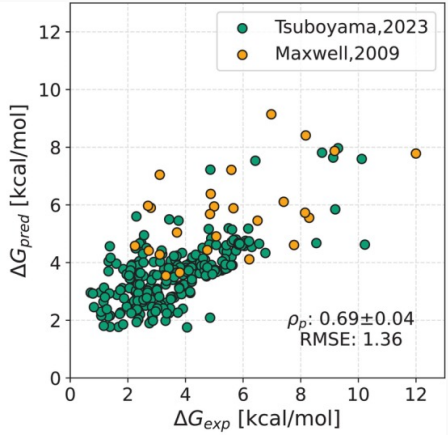


Summed native-residue probabilities correlate with ΔG ($r = 0.69$), more strongly than contact count or chain length.

What this establishes: useful stability information beyond a simple size proxy. It does not establish that every model score is a free energy.

Paper Fig. 1a–c. Orange: Maxwell; green: Tsuboyama. Axes and bootstrap errors retained.

Calibration yields about 1.4 kcal/mol fitted RMSE



265 proteins

Pearson $r = 0.69$
RMSE $\simeq$ 1.4 kcal/mol

The linearly rescaled score captures a broad trend across unrelated proteins.

These points also supplied the two calibration parameters; distinguish fitted performance from an independently held-out test.

Paper Fig. 2 labels RMSE 1.36; Table 1 gives 1.39 kcal/mol. Abstract summarizes error as about 1.5.

The gain is useful, but the simple baselines matter

Calibrated RMSE (kcal/mol)	
Null model	1.88
Length model	1.55
Contact-count model	1.53
ESM-IF	**1.39**

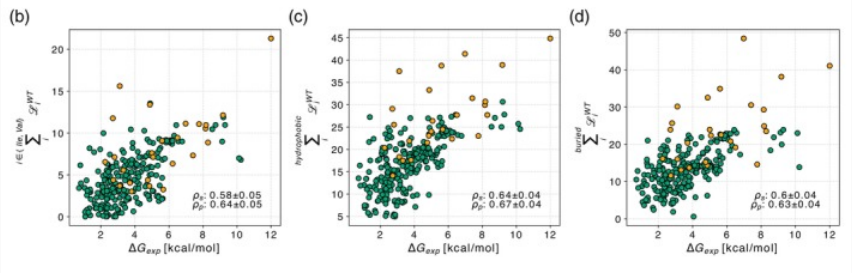
Alternative score correlations	
ESM-1b	0.25
FoldX	0.04
Chroma	0.62
ESM-IF	**0.69**

ESM-IF improves fitted RMSE by 0.14 kcal/mol over contact count. Chroma also carries substantial signal.

FoldX was developed mainly for mutation effects; this comparison does not invalidate that use. ESM-1b's weak correlation applies to this scoring recipe.

Tables 1–2; Fig. S5 as discussed in Results 2.1. ACN: Cα contact cutoff 8 Å.

Hydrophobic residue scores carry much of the signal



Valine + isoleucine: $r = 0.64$; all hydrophobic residues: $r = 0.67$; buried residues: $r = 0.63$; all residues: $r = 0.69$.

Author interpretation: consistent with the hydrophobic effect.

Our caution: correlation of selected score components is not a causal energy decomposition.

Paper Fig. 3b–d (cropped); Results 2.1. Backbone-only representation still predicts residue compatibility.

How strong is the evidence for calibration generalization?

Paper reports: errors remain robust under cross-validation (Fig. S4); error is mostly constant across stability values (Fig. S3).

Reproducibility caveat from the public SI notebook:

1. The leave-one-out cell calculates fitted parameters named `popt_cv`.
2. It then assigns from `popt` and predicts using the globally fitted `a_if, b_if`.

As written, this cell does not demonstrate genuinely held-out calibration. This is a code-reading observation, not a rerun or a claim about all analyses.

Useful follow-up: rerun calibration with held-out parameters and sequence/fold-separated splits.

Author repository, paper_SI/dG_pred_SI.ipynb, "cross-validation fitting" cell; accessed 2026-11-01.

Can two absolute predictions reveal a conformational gap?

For the *same sequence*, assume states G and E share an unfolded reference U:

$$\Delta G_{G\to U} = G_U - G_G, \qquad \Delta G_{E\to U} = G_U - G_E.$$

Subtracting cancels U:

$$G_E - G_G = \Delta G_{G\to U} - \Delta G_{E\to U}.$$

Score the same sequence against two backbone structures; subtract their calibrated stabilities.

Requirement: the structures must represent the experimentally observed states. A single backbone cannot capture every side-chain or ensemble contribution.

Results 2.2; thermodynamic cycle in Fig. 4c. This is a test of extrapolation, not part of the 265-protein fit.

Two state-pair tests get the sign, underestimate the gap

System	Experiment	Prediction
	E relative to G, kcal/mol	
T4 lysozyme L99A	∼2.0	1.2
Cyclophilin A	1.7	0.3

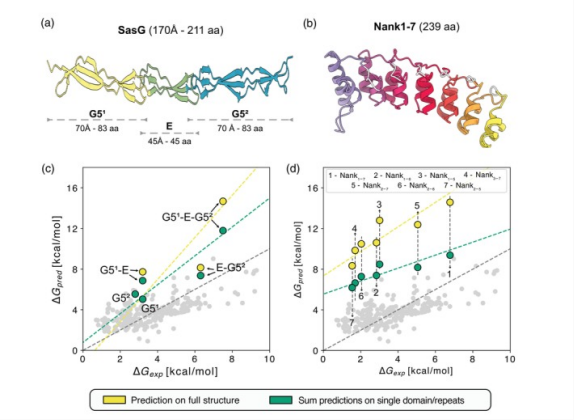
Both E states are correctly less stable. Quantitative gaps remain inaccurate.

Cyclophilin A backbones differ by only 0.34 Å RMSD; how fully these crystal structures represent the NMR exchange remains uncertain.

The gaps are comparable to the absolute-prediction error; two successes in sign do not establish general conformational thermodynamics.

Paper Fig. 4a–e; Results 2.2. T4L: 164 aa; Cyclophilin A: 165 aa, slightly beyond the main size range.

Multidomain systems expose a size and cooperativity problem



SasG and Notch ankyrin repeats

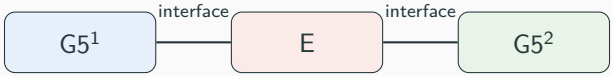
Full-construct scoring (yellow) overestimates stability.

An additive treatment of individual repeats/domains (green) moves predictions closer to experiment.

But the stability gained by adding repeats remains inaccurate.

Paper Fig. 5a–d; Results 2.3; Methods 4.4.5. Gray points: the original small-protein benchmark.

Why adding scores is not enough for cooperative folding



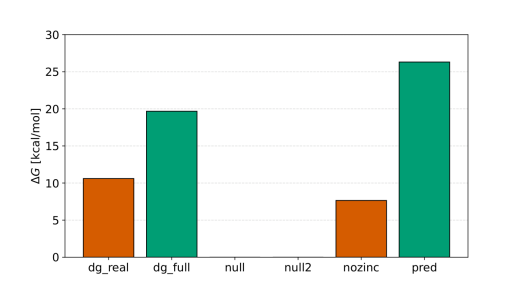
SasG's E domain is disordered by itself and folds in the appropriate neighboring-domain context. A fixed folded E-domain input can assume the very stability being assessed.

Separately scoring domains omits important interface contributions. Whole-chain scoring also does not explicitly represent alternative unfolding pathways.

Lesson: sequence–backbone compatibility does not by itself identify a system's thermodynamic states.

Results 2.3; Gruszka et al. (2012, 2015), as analyzed by the paper. Schematic adapted for teaching.

Multistate folding and large proteins are outside the safe scope



Two monomeric, reversible multistate systems were examined: creatine kinase and exo-small β-lactamase. Both are overestimated.

Excluded large protein VlsE: predicted 18 versus experimental 5.7 kcal/mol.

These failures argue against extrapolating the fitted relation across protein size and folding complexity.

Results 2.3; Methods 4.1; author SI repository figure figsi_3states_prots (corresponding to Fig. S6).

What a practical use of this method would require

1. **Specify the physical target.** Approximate ambient-condition, two-state stability of a small natural protein.
2. **Prepare the structure and sequence carefully.** Paper uses AF2/LocalColabFold models; templates and reduced MSA depth retain alternative conformations when needed.
3. **Score and calibrate consistently.** Match probability normalization and native-sequence conditioning; report kcal/mol and model version.

Use as a screening estimate with roughly 1.4–1.5 kcal/mol errors in the studied regime. A point prediction alone is not a protein-specific uncertainty estimate.

Code and a Colab implementation: KULL-Centre/_2024_cagiada_stability.

Methods 4.2–4.3; Data availability. Practical interpretation, not an additional validation result.

Discussion: what would make the result convincing at scale?

1. **Generalization:** does the gain over contacts survive correct calibration CV, fold-separated tests, and an independent experimental platform?
2. **Physical representation:** do side-chain-aware or ensemble models improve conformational gaps and multidomain cooperativity?
3. **Experimental coverage:** what new consistent-condition ΔG data are needed beyond short natural two-state folders?

Take-home: an unexpectedly simple readout of a design model predicts absolute stability with useful accuracy in a narrow regime. Its extrapolation failures reveal where sequence–structure compatibility falls short of thermodynamics.

Conclusions and limitations; discussion questions are our interpretation.

Sources and reading guide

Primary article

Cagiada M, Ovchinnikov S, Lindorff-Larsen K. *Protein Science* 34(1):e5233 (2025). Online publication: December 14, 2024.
doi:10.1002/pro.5233 Full journal text: PMC11645669

Reproducibility material

Author code, data, and SI analysis notebook. Main-paper figures reproduced from the journal text; additional SI plot from author repository.

Access qualification

Full article and relevant author SI code/plots were read. The journal's supplementary PDF could not be retrieved; its additional prose was not independently checked. The cross-validation caveat concerns the public notebook as accessed on November 1, 2026.